Filesystem & Storage

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File system:-

A file system is the structure used by the OS, to store & organize data on a disk.

without file system [Disk = random bits (0s & 1s)]
with file system
Disk = organized files + folders.

So file system = organization + manage.

File ⇒ A file is a collection of data stored on disk.

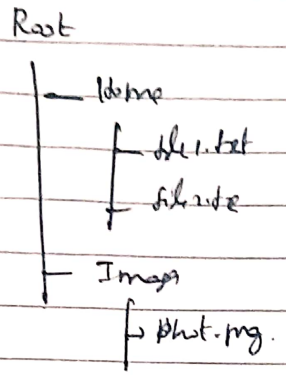
Ex
- .txt → These are file extensions → text
- .jpg → image file
- .png → image
- .mp4 → video file
- .exe → executable file.

File = Data + Metadata
Data ↓ Actual data bit
Metadata ↓ data about data
⇓
☆☆☆ Metadata includes ⇒
- File name
- size
- permission
- owner
- Creation time
- location on disk

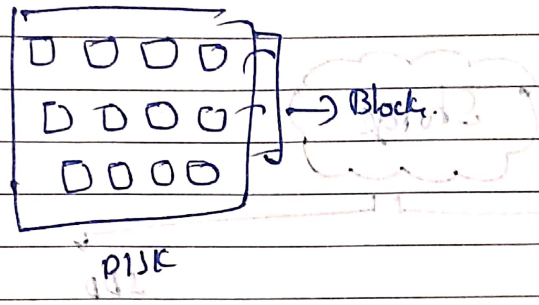
{ So file has actual data & meta data }.

Directory [Folder structure]:

Directory organizes the files in a tree structure



An file will stored internally on the blocks in the disk..



A file is stored in one (or) more blocks...

In which we have algorithm

- continuous file structure
- Index based file storage
- Linked blocks

AA

OS finds a file → by the

when user click on a file → open("file.txt")

which block contain such B

File system Types

Windows → NTFS
FAT32
exFAT

LINUX → ext4
ext 3

MAC → APFS

File operation → Create
Read
write
append
Delete
open
close

Storage

HDD
(Hard disk Drive)
- Mechanical (spinning disk)
- Slower
- cheaper

SDD
(Solid-state Drive)
- No moving parts
- Faster
- More expensive.

* How HDD works →

HDD has:
- platter (spinning disks)
- Read / write head.

Stem
- Head moves to correct track (Seek Time)
- Disk rotates
- Data read.

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SSD works

SSD uses Flash memory

Direct access to memory cell..

$$\boxed{\text{Disk Access time} = \text{Seek Time} + \text{Rotational Delay} + \text{Transfer time}}$$

SSD = Faster Direct access..

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